

Systems Thinking Project

Acrobot

Phase 3: Control Around the Equilibria

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November 11, 2026

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1 Introduction

1.1 What is an Acrobot?

An acrobot is a two-link pendulum that hangs from a fixed pivot. The first link (the "shoulder" link) is free to rotate about the pivot; the second link is joined to the end of the first at the "elbow". The only motor sits at the elbow, so the shoulder is *passive*. It is the robot version of a gymnast on a high bar who can only swing the legs at the hips. Because there are two degrees of freedom but only one actuator, the acrobot is a standard example of an *underactuated* system [1], [2].

1.2 Model

The configuration is described by two generalised coordinates: q_1, the angle of link 1 from the downward vertical, and q_2, the angle of link 2 relative to link 1. The full state contains the angles and their rates:

$$\mathbf{q} = \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} q_1 \\ q_2 \\ \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}.$$

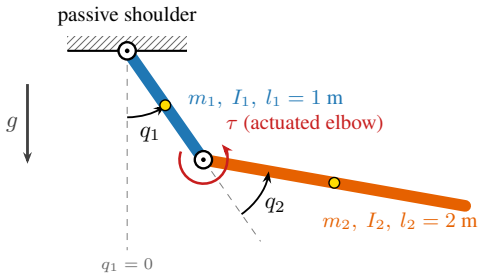


Figure 1: The acrobot: passive shoulder at the pivot, motor torque τ at the elbow; dots mark the centres of mass.

1.3 Equations of Motion

The dynamics (given in the Phase 1 helper and verified in Phase 1) are

$$M(q)\,\ddot{q} + C(q,\dot{q})\,\dot{q} = \tau_g(q) + B\,\tau, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad |\tau| \le 20\,\mathrm{N\,m}. \tag{1}$$

τ **(Input Torque):** the torque of the elbow motor, the only quantity we command. $B = [0,\ 1]^T$ says it acts on the elbow equation only.

$M(q)$ **(Inertia Matrix):** the resistance to angular acceleration. It depends on the elbow angle because the mass distribution changes as the arm folds:

$$M(q) = \begin{bmatrix} 2.663 + 2\cos q_2 & 1.33 + \cos q_2 \\ 1.33 + \cos q_2 & 1.33 \end{bmatrix}, \qquad \det M = 1.7729 - \cos^2 q_2 \ge 0.7729 > 0. \tag{2}$$

$C(q,\dot{q})$ **(Coriolis Matrix):** the coupling forces between the links (centrifugal and Coriolis effects):

$$C(q,\dot{q}) = \sin q_2 \begin{bmatrix} -2\dot{q}_2 & -\dot{q}_2 \\ \dot{q}_1 & 0 \end{bmatrix}. \tag{3}$$

$\tau_g(q)$ **(Gravity Torque):** the torque that gravity exerts on each joint, $\tau_g = -\partial U/\partial q$, with potential energy U:

$$U = -14.715\cos q_1 - 9.81\cos(q_1+q_2), \qquad \tau_g(q) = -9.81\begin{bmatrix} 1.5\sin q_1 + \sin(q_1+q_2) \\ \sin(q_1+q_2) \end{bmatrix}. \tag{4}$$

System Parameters:

$$m_1 = m_2 = 1\,\mathrm{kg}, \qquad l_1 = 1\,\mathrm{m}, \qquad l_2 = 2\,\mathrm{m}, \qquad l_{c1} = 0.5\,\mathrm{m}, \qquad l_{c2} = 1\,\mathrm{m},$$
$$I_1 = 0.083\,\mathrm{kg\,m^2}, \qquad I_2 = 0.33\,\mathrm{kg\,m^2}, \qquad g = 9.81\,\mathrm{m\,s^{-2}}.$$

1.4 Equilibria and Linear Model (from Phase 1)

With $\tau^* = 0$ the equilibria satisfy $\tau_g(q^*) = 0$, i.e. $\sin q_1^* = 0$ and $\sin(q_1^* + q_2^*) = 0$. This gives four postures:

Table 1: The four equilibria

	Posture	(q_1^*, q_2^*)	U [J]	Description
E_1	down–down	$(0, 0)$	-24.525	both links hanging
E_2	down–up	$(0, \pi)$	-4.905	link 1 down, link 2 up
E_3	up–up	$(\pi, 0)$	$+24.525$	fully upright (target)
E_4	up–down	(π, π)	$+4.905$	link 1 up, link 2 down

Linearizing about each equilibrium with $\tilde{x} = x - x^*$ gives

$$\dot{\tilde{x}} = A\tilde{x} + B_\ell\tilde{\tau}, \qquad A = \begin{bmatrix} 0 & I \\ P & 0 \end{bmatrix}, \qquad B_\ell = \begin{bmatrix} 0 \\ r \end{bmatrix}, \tag{5}$$

with $P = M(q^*)^{-1}K(q^*)$ and $r = M(q^*)^{-1}B$, where $K = \partial\tau_g/\partial q$. The transfer functions from τ to the angles are

$$G_1(s) = \frac{r_1 s^2 + (P_{12}r_2 - P_{22}r_1)}{\Delta(s)}, \qquad G_2(s) = \frac{r_2 s^2 + (P_{21}r_1 - P_{11}r_2)}{\Delta(s)}, \tag{6}$$
$$\Delta(s) = s^4 - \mathrm{tr}(P)\,s^2 + \det P.$$

Table 2: Transfer functions at the four equilibria (Phase 1)

	$\Delta(s)$	$N_1(s)$ $(\tau \to q_1)$	$N_2(s)$ $(\tau \to q_2)$
E_1	$s^4 + 42.2410\,s^2 + 186.7719$	$-3.0147\,(s^2 + 4.2103)$	$6.0332\,(s^2 + 5.2595)$
E_2	$s^4 + 8.4025\,s^2 - 186.7719$	$-0.4270\,(s^2 - 29.7273)$	$0.8578\,(s^2 + 7.3982)$
E_3	$s^4 - 42.2410\,s^2 + 186.7719$	$-3.0147\,(s^2 - 4.2103)$	$6.0332\,(s^2 - 5.2595)$
E_4	$s^4 - 8.4025\,s^2 - 186.7719$	$-0.4270\,(s^2 + 29.7273)$	$0.8578\,(s^2 - 7.3982)$

1.5 Objective

Design controllers that stabilize the acrobot at the upright equilibrium E_3 and at least one other equilibrium, and justify the design choices. Characterize the limits of each controller: how far from the equilibrium it still works with $\tau_{\max} = 20\,\mathrm{N\,m}$.

Phase 2 showed that E_1 is only marginally stable, that E_2, E_3, E_4 are unstable (1, 2 and 1 RHP poles), and that one measured angle cannot stabilize E_3 with PD feedback. That result guides the design here.

2 Control Around the Equilibria

2.1 Controller Design

The rank of $\mathcal{C}$ is 4 at every equilibrium, so all four states can be fed back. Including the torque limit, every controller has the form

$$\tau = \mathrm{sat}(-K\tilde{x}), \qquad \tilde{x} = [\tilde{q}_1,\ \tilde{q}_2,\ \dot{q}_1,\ \dot{q}_2]^T, \qquad |\tau| \le 20\,\mathrm{N\,m}. \tag{7}$$

The controllers used are:

- **LQR** at the upright E_3, at E_2 and at E_4;
- **pole placement** at E_3, for comparison;
- **PD on the elbow angle only** at E_1.

2.2 Linear Quadratic Regulator (LQR)

K minimizes a cost that penalizes both the state error and the torque:

$$J = \int_0^\infty \left(\tilde{x}^T Q \tilde{x} + R\,\tau^2\right) dt, \qquad A^T S + S A - S B_\ell R^{-1} B_\ell^T S + Q = 0, \qquad K = R^{-1} B_\ell^T S. \tag{8}$$

Expected performance of LQR:

- **Stability:** guaranteed for any $Q \succeq 0$, $R > 0$ when (A, B_ℓ) is controllable.
- **Robustness:** phase margin $\ge 60^\circ$ and gain margin $[\frac{1}{2}, \infty)$ [7].
- **Tuning:** $Q = I$ weighs 1 rad of angle like 1 rad/s of rate. A smaller R makes the controller faster but demands more torque, so R trades speed against saturation.

2.2.1 Choice of R

For each R we simulated the nonlinear saturated loop from a grid of initial offsets and counted how many were recovered (Fig. 2).

- **E_3:** the recovered area stays at 16–20% of a $\pm 60^\circ$ grid for $10^{-3} \le R \le 1$, while the fastest pole moves from -10 to $-214\ \mathrm{s}^{-1}$. The limit is $\tau_{\max}$, not the tuning. $R = 0.03$ is the knee: 93% of the best area with a fastest pole of -40.

- E_2: the region collapses for $R > 0.3$ ($99\% \to 90\% \to 20\%$), so $R = 0.1$.
- E_4: the area peaks at $R \approx 0.1$–0.18, so $R = 0.1$.

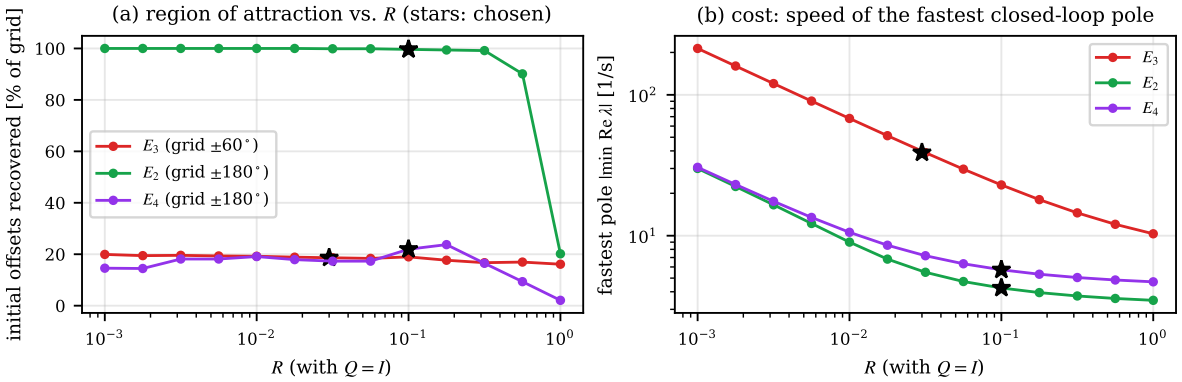


Figure 2: Weight selection ($Q = I$): (a) share of offsets recovered; (b) fastest closed-loop pole. Stars: chosen R.

2.3 Pole Placement at E_3 (Comparison)

Placing the poles at $\{-3, -4, -5, -6\}$ gives $K = [-322.9, -134.6, -138.1, -66.0]$. It settles faster from 2° (1.3 s vs 4.1 s) and with less torque (11.3 vs 17.6 N m), but recovers a smaller region (20% vs 26% of ±40°; 5° vs 6° along q_1). LQR is therefore kept for E_3.

2.4 E_1: PD on the Elbow Angle Only

At E_1 the zeros of G_2 are imaginary ($\pm j2.29$), so the PD sign conflict found at E_3 in Phase 2 does not arise and one angle suffices:

$$\tau = -K_p\tilde{q}_2 - K_d\dot{q}_2, \qquad K_p = -5, \quad K_d = 0.4, \tag{9}$$

the pair with the best decay rate on a (K_p, K_d) grid ($\max \mathrm{Re}\,\lambda = -0.52$). **Proof of stability (energy shaping, LaSalle [4]):**

$$V_c = E - E^* + \tfrac{1}{2}K_p\tilde{q}_2^2, \qquad \dot{V}_c = -K_d\,\dot{q}_2^2 \le 0, \tag{10}$$

$$\nabla^2\left(U + \tfrac{1}{2}K_p\tilde{q}_2^2\right)\Big|_{E_1} \text{ has eigenvalues } 0.760,\ 28.575 > 0.$$

V_c has a strict minimum at E_1 and never increases, and $\dot{q}_2 \equiv 0$ only at E_1, so E_1 is asymptotically stable. Pure damping ($K_p = 0$) decays 19× more slowly.

2.5 Designed Controllers

Table 3: Phase 3 controllers and their closed-loop poles

Controller	Design	$K = [k_{q_1}, k_{q_2}, k_{\dot{q}_1}, k_{\dot{q}_2}]$	Closed-loop poles
E_3 LQR	$Q = I$, $R = 0.03$	$[-503.0, -203.9, -218.3, -101.5]$	-39.86, $-2.25 \pm 0.10j$, -1.34
E_3 pole placement	$\{-3, -4, -5, -6\}$	$[-322.9, -134.6, -138.1, -66.0]$	-6, -5, -4, -3
E_2 LQR	$Q = I$, $R = 0.1$	$[6.60, 46.49, 5.23, 13.24]$	-4.26, -2.40, $-1.24 \pm 4.16j$
E_4 LQR	$Q = I$, $R = 0.1$	$[-49.31, 38.91, -13.19, 6.10]$	-5.73, -3.21, $-0.96 \pm 3.08j$
E_1 elbow PD	$K_p = -5$, $K_d = 0.4$	$[0, -5, 0, 0.4]$	$-0.69 \pm 2.31j$, $-0.52 \pm 2.14j$

2.6 Limits with $\tau_{\max} = 20\,\mathrm{N\,m}$

2.6.1 Guaranteed Region

Inside $\Omega_c = \{\tilde{x}^T S \tilde{x} \le c\}$ the command never saturates if

$$c^* = \frac{\tau_{\max}^2}{K S^{-1} K^T}, \qquad \dot{V} = -\tilde{x}^T (Q + K^T R K)\tilde{x} < 0 \ \text{ on } \Omega_{c^*}. \tag{11}$$

At rest this guarantees $0.72°/1.66°$ (E_3), $45.6°/19.9°$ (E_2) and $20.0°/23.1°$ (E_4) along q_1/q_2. The bound is conservative because short saturation is usually survived.

2.6.2 Simulated Region of Attraction

The real limits come from simulating the nonlinear saturated loop from a grid of initial offsets at rest. A start counts as recovered if it ends within 1° and 0.05 rad/s of q^*. The fast integrator used for the grids matches PyBullet on 100% of random points and 96% of boundary points.

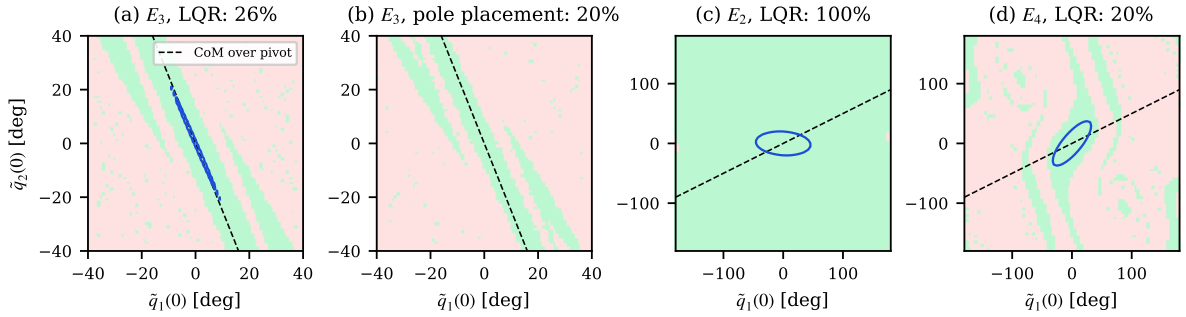


Figure 3: Region of attraction (green: recovered). Blue ellipse: guaranteed region. Dashed: offsets with the centre of mass over the pivot.

Table 4: How far from the equilibrium each controller still works

Controller	Recovered (grid)	$\lvert\tilde{q}_1(0)\rvert$ alone	$\lvert\tilde{q}_2(0)\rvert$ alone	Shoulder kick	Settling
E_3 LQR	26% (±40°)	6°	13°	0.20 rad/s	4.1 s
E_3 pole placement	20% (±40°)	5°	9°	0.15 rad/s	1.3 s
E_2 LQR	99.7% (±180°)	175°	180°	≥ 8 rad/s	3.0 s
E_4 LQR	20.5% (±180°)	35°	55°	2.1 rad/s	3.6 s
E_1 elbow PD	77.8% (±180°)	40°	115°	2.15 rad/s	12.8 s

Why the upright region is a thin band. Linearizing τ_g at E_3:

$$\tau_{g,1} = 9.81\,(2.5\,\tilde{q}_1 + \tilde{q}_2). \tag{12}$$

Gravity pushes the passive shoulder in proportion to the horizontal offset of the total centre of mass. The motor can only counter this through the elbow coupling. Along $\tilde{q}_2 = -2.5\,\tilde{q}_1$ there is no shoulder torque, and the region reaches the grid edge ($\ge 43°$). Across that line, 20 N m is used up after about 6°.

2.7 Time Responses

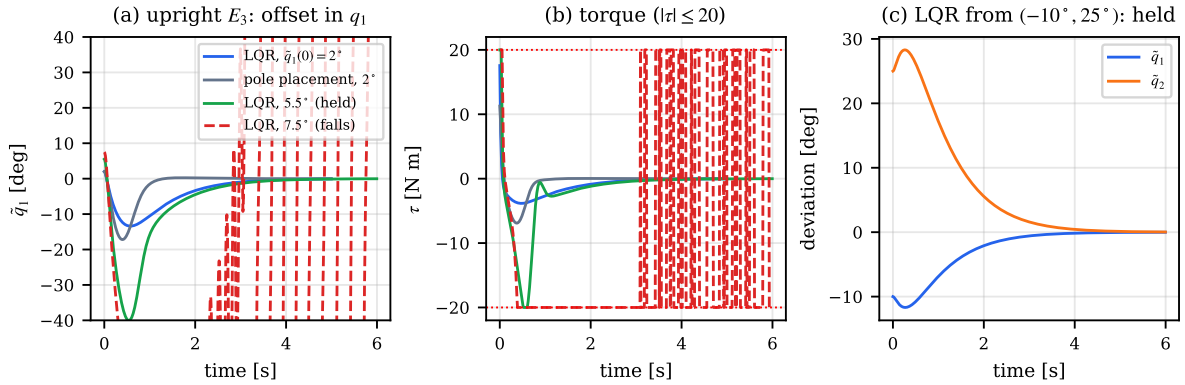


Figure 4: E_3 in PyBullet: LQR is held from 5.5° and falls from 7.5°; a CoM-balanced offset $(-10°, 25°)$ is held with only 3.3 N m.

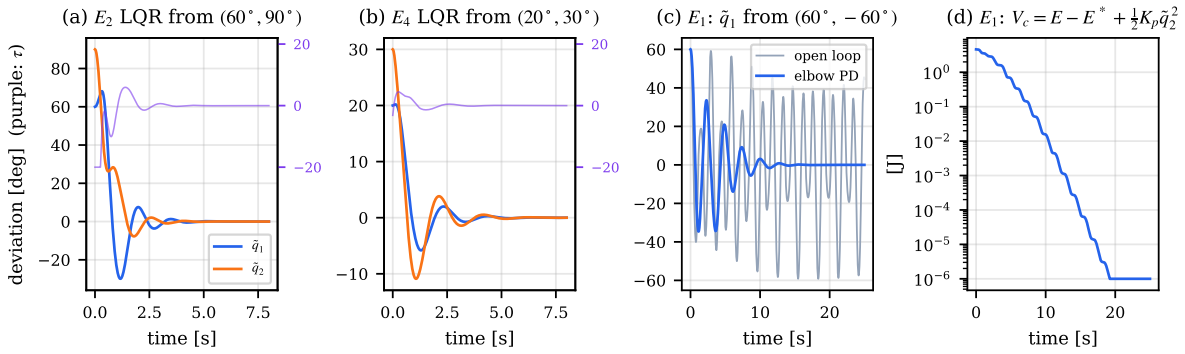


Figure 5: (a) E_2 and (b) E_4 with LQR; (c) E_1 with elbow PD vs. open loop; (d) V_c decreasing along the E_1 run.

Observations:

- **Settling time:** pole placement settles fastest at E_3, but LQR rescues a larger region. E_1 is slowest (12.8 s) because its slow mode barely moves the elbow.
- **Control effort:** at E_3 even a 2° offset needs 17.6 N m, close to the limit, which explains the small region. At E_4 the peak is only 4.5 N m.
- **Steady-state error:** zero for every controller (each converges to q^* within 0.001°).

3 Simulations and Code

3.1 PyBullet Simulator

All simulations use the full nonlinear model in PyBullet with an interactive dashboard (Fig. 6). The user picks an equilibrium, an initial condition (slider, typed value or by

dragging the arm) and a controller: open loop, full-state LQR, pole placement, elbow PD (for E_1), or the Phase 2 single-angle controllers. The dashboard shows $\tilde{q}(t)$, $\tau(t)$ and the closed-loop poles live. File: `phase3/acrobot_phase3_sim.py`; instructions in `phase3/README.md`.

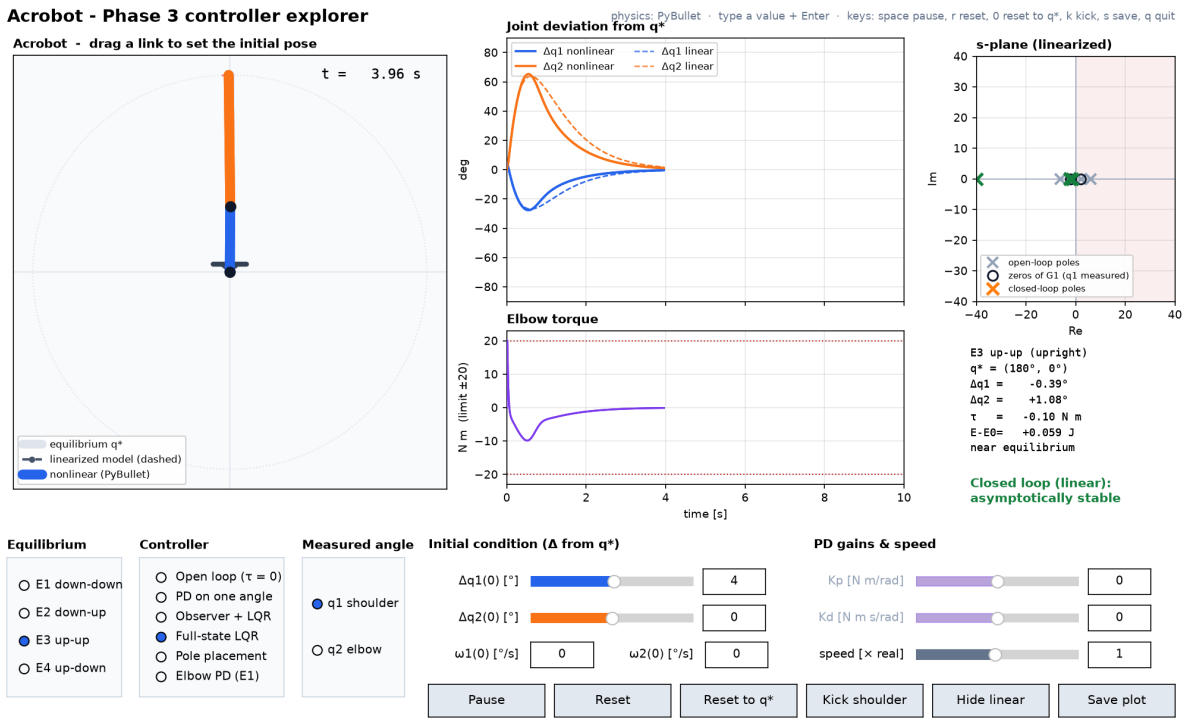


Figure 6: Phase 3 dashboard: full-state LQR catching the upright from 4°.

3.2 MATLAB Codes

It designs every controller (`lqr`, `place`), prints the gains, poles and guaranteed regions, runs the nonlinear saturated loops with `ode45`, and maps the regions of attraction. Gains, poles and guaranteed regions match the tables above exactly.

The files are listed below; the algorithms they implement follow as pseudo-code.

Table 5: MATLAB files

File	Purpose
`Acrobot_Nonlinear_Dynamics.m`	nonlinear model $\dot{x} = f(x, \tau)$ with the 20 N m limit (Phase 1)
`Acrobot_Linearize.m`	A, B_ℓ at an equilibrium (Phase 1)
`Acrobot_Phase3_Design.m`	the controllers (LQR, pole placement, elbow PD)
`Acrobot_Phase3_Main.m`	guaranteed regions, nonlinear runs, region-of-attraction maps

Algorithm 1 Nonlinear dynamics $\dot{x} = f(x, \tau)$ (`Acrobot_Nonlinear_Dynamics.m`) and linearization (`Acrobot_Linearize.m`)

function DYNAMICS($x = [q_1, q_2, \dot{q}_1, \dot{q}_2]$, τ)
  $\tau \leftarrow \max(-20, \min(20, \tau))$ ▷ motor limit
  build $M(q)$, $C(q, \dot{q})$ and $\tau_g(q)$ from the given model
  solve $M\,\ddot{q} = \tau_g + B\tau - C\dot{q}$ for $\ddot{q}$ ▷ backslash, never M^{-1}
  return $[\dot{q};\ \ddot{q}]$
end function
function LINEARIZE(q^*)
  $K \leftarrow \partial\tau_g/\partial q$ at q^*; $\quad P \leftarrow M(q^*)^{-1}K$; $\quad r \leftarrow M(q^*)^{-1}B$
  return $A = \begin{bmatrix} 0 & I \\ P & 0 \end{bmatrix}$, $B_\ell = \begin{bmatrix} 0 \\ r \end{bmatrix}$
end function

Algorithm 2 Controller design and limits (`Acrobot_Phase3_Design.m`, `Acrobot_Phase3_Main.m`)

for each design (LQR at E_2, E_3, E_4; pole placement at E_3; elbow PD at E_1) **do**
  $(A, B_\ell) \leftarrow$ LINEARIZE(q^*)
  if LQR **then**
    $(K, S) \leftarrow$ `lqr`$(A, B_\ell, Q = I, R)$; $\quad c^* \leftarrow \tau_{\max}^2/(KS^{-1}K^T)$ ▷ guaranteed region
  else if pole placement **then**
    $K \leftarrow$ `place`$(A, B_\ell, \{-3, -4, -5, -6\})$
  else
    $K \leftarrow [0,\ K_p,\ 0,\ K_d]$ ▷ elbow angle only
  end if
  print K and eig$(A - B_\ell K)$
end for
function REGIONOFATTRACTION(K, q^*, grid of $(\tilde{q}_1(0), \tilde{q}_2(0))$)
  for each grid point **do**
    simulate $\tau = \text{sat}(-K\tilde{x})$ for 10 s from rest ▷ nonlinear model
    mark it recovered if it ends within 1° and 0.05 rad/s of q^*
  end for
  return the map, the share recovered and the largest offset along each axis
end function

3.3 Animation

A 2K Manim animation of this phase: watch the video.

4 Tabulation

Table 6: Summary of the Phase 3 controllers ($|\tau| \leq 20\,\mathrm{N\,m}$)

Controller	Design	Recovered (grid)	Along q_1 / q_2 alone	Guaranteed
E_3 LQR	$Q = I$, $R = 0.03$	26% ($\pm 40°$)	6° / 13°	0.72° / 1.66°
E_3 pole placement	$\{-3, -4, -5, -6\}$	20% ($\pm 40°$)	5° / 9°	–
E_2 LQR	$Q = I$, $R = 0.1$	99.7% ($\pm 180°$)	175° / 180°	45.6° / 19.9°
E_4 LQR	$Q = I$, $R = 0.1$	20.5% ($\pm 180°$)	35° / 55°	20.0° / 23.1°
E_1 elbow PD	$K_p = -5$, $K_d = 0.4$	77.8% ($\pm 180°$)	40° / 115°	–

5 Conclusion

All four equilibria were stabilized. LQR was chosen for E_3, E_2 and E_4 because it is guaranteed stable with good margins and trades speed against torque with one weight R. The weight was chosen by measuring the region of attraction of the nonlinear saturated loop. At the upright that region hardly depends on R: the $20\,\mathrm{N\,m}$ limit, not the tuning, restricts LQR to 6° in q_1 or 13° in q_2, and to a thin band along the line where the centre of mass is over the pivot, because gravity twists the motorless shoulder in proportion to that offset. Pole placement settles faster but rescues less. E_2 is recovered from almost any rest position and E_4 from about 35°/55°. E_1 is stabilized from the elbow angle alone, with an energy-shaping Lyapunov proof.

6 Contributions

1. Report compilation – _____
2. Controller design (Sections 2.1–2.5) – _____
3. Limits and region of attraction (Section 2.6) – _____
4. Time responses (Section 2.7) – _____
5. PyBullet simulator and figures – _____
6. MATLAB codes – _____
7. Animation (Manim) – _____

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